

Indian Road Accident Analysis

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Role: Data Analyst Project

Tools Used: Python, Pandas, NumPy, Matplotlib, Seaborn, Tableau

Project Overview

This project presents an end-to-end analysis of **20,000 Indian road accident records** using **Python, Tableau, and Business Intelligence techniques**. The objective was to identify accident patterns, evaluate road safety risks, and generate actionable business insights through data analysis and visualization. The project followed a structured workflow including data cleaning, exploratory data analysis, business analysis, advanced analytics, dashboard development, and business reporting.

Tools & Technologies

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Tableau
- Git & GitHub

Key Project Achievements

- Analyzed **20,000** road accident records.
- Performed complete **data cleaning and preprocessing**.
- Conducted **Exploratory Data Analysis (EDA)**.
- Solved **50+ business analysis questions**.

- Developed custom analytical metrics including **Risk Hotspot Score**, **Safety Score**, **Cause Impact Score**, **Weather Risk Index**, **Peak Hour Severity Index**, and **Executive Priority Score**.
- Built an interactive **Tableau dashboard** for executive reporting.
- Produced business recommendations based on analytical findings.

Major Findings

- Maharashtra recorded the highest number of accidents.
- Karnataka reported the highest fatality rate.
- Pune recorded the highest number of fatal accidents.
- Overspeeding was identified as the most impactful accident cause.
- High-risk accident groups showed higher severity and casualty levels.

Business Value

The project demonstrates how data analytics can transform raw accident records into meaningful business intelligence. The insights can support transportation authorities, policymakers, and urban planners in improving road safety, prioritizing infrastructure investments, and allocating resources more effectively.

Skills Demonstrated

- Data Cleaning & Preprocessing
- Exploratory Data Analysis (EDA)
- Business Analytics
- SQL Analysis
- KPI Development
- Data Visualization

- Dashboard Development
- Business Reporting
- Problem Solving

Conclusion

This project demonstrates how data analytics can transform large volumes of accident records into meaningful business insights.

By combining exploratory data analysis, business intelligence techniques, feature engineering, advanced analytics, and interactive visualization, the analysis provides a comprehensive understanding of accident patterns across different geographical regions, environmental conditions, and road characteristics.

The findings can support government agencies, transportation authorities, and urban planners in identifying high-risk areas, prioritizing safety investments, improving infrastructure planning, and developing targeted accident prevention strategies.

The project also establishes a strong foundation for future work, including interactive dashboards, machine learning models for accident severity prediction, and real-time road safety monitoring systems.